MS^{2}AE-Net: A Multiscale Spectral–Spatial Autoencoder Network for Hyperspectral Unmixing

Ruyi Feng, *Senior Member, IEEE*, Hangqi Peng, *Student Member, IEEE*, and Lizhe Wang, *Fellow, IEEE*

***Abstract*—Hyperspectral unmixing (HU) is a critical technique for analyzing the composition of materials within remotely sensed scenes. However, the inherent presence of mixed pixels and the challenge of comprehensively utilizing the rich spectral–spatial information in hyperspectral images (HSIs) limit the accuracy of existing methods. Many approaches have focused on multiscale spatial features but often underestimate the significant potential of multiscale spectral information, which is crucial for learning a comprehensive and robust spectral representation. This article proposes a multiscale spectral–spatial autoencoder network (MS^{2}AE-Net) for HU, which features two novel modules designed for comprehensive multiscale feature extraction. MS^{2}AE-Net uniquely integrates two specialized modules within an autoencoder (AE) framework: a multiscale spectral module (MSpecM) and a multiscale spatial module (MSpatM). The MSpecM captures spectral characteristics at various resolutions by first decomposing spectral signals into low-frequency (invariant) and high-frequency (variant) components using fast Fourier transform (FFT) and then employing adaptive top-down and bottom-up fusion strategies within its spectral hierarchical fusion block (SHFB). Concurrently, the MSpatM, built upon scale-aware spatial blocks (SASBs) that utilize the scale-aware modulation (SAM) concept, effectively extracts multiscale spatial features through multihead mixed convolution (MHMC) and aggregates them using scale-aware aggregation (SAA). By simultaneously learning from multiple scales in both spectral and spatial domains, MS^{2}AE-Net generates more robust and discriminative feature representations. Experimental results on synthetic and real-world datasets demonstrate the superiority of the proposed MS^{2}AE-Net over several state-of-the-art unmixing methods, particularly in terms of abundance estimation accuracy and robustness to noise.**

***Index Terms*—Autoencoder (AE), hyperspectral unmixing (HU), multiscale, spatial–spectral.**

I. Introduction

HYPERSPECTRAL imaging technology, with its capability to capture rich spectral information across hundreds of contiguous narrow bands, has emerged as a powerful tool for detailed analysis of materials and scenes. This technology finds widespread applications in diverse fields, including but not limited to precision agriculture [1], forestry [2], environmental monitoring [3], and mineral exploration [4]. However, a significant challenge in hyperspectral image (HSI) analysis arises from the presence of mixed pixels. Due to limited spatial resolution, a single pixel often covers multiple distinct ground objects, resulting in a mixed spectral signature that severely confounds accurate identification and quantitative analysis of these components, thereby greatly limiting the accuracy and reliability of downstream applications [5]. To address this critical issue, hyperspectral unmixing (HU) technology has been developed. The core objective of HU is to decompose the mixed spectrum of a pixel into a set of pure constituent spectra, known as endmembers, and their corresponding proportions within the pixel, referred to as abundances, thus enabling subpixel-level compositional analysis of the scene [6].

Effectively performing HU necessitates the establishment of a mathematical model that adheres to certain physical assumptions regarding the light interaction within a mixed pixel. Based on these assumptions, mixing models can be broadly categorized into linear mixing models (LMMs) and nonlinear mixing models (NLMMs) [7]. LMM assumes that the light captured by the sensor from a mixed pixel undergoes only a single reflection from the surface of each constituent material before reaching the sensor. In this model, the observed spectrum is a linear combination of the endmember spectra, weighted by their respective abundances [8]. In contrast, NLMM are designed to account for more complex light scattering processes, such as multiple reflections [9], [10], [11], [12] and intimate mixtures [13], which lead to nonlinear relationships between endmember spectra and the observed mixed spectrum. Due to its simplicity and computational efficiency, LMM has been widely adopted in HU.

Based on the objective of HU, the problem can be further decomposed into three interconnected subproblems: determining the number of endmembers, extracting the endmember spectra, and estimating the abundances. Estimating the number of endmembers is a crucial initial step, and feature analysis-based algorithms, such as HySime [14] and virtual dimension (VD) [15], have been widely employed for this purpose. For endmember extraction, existing endmember extraction algorithms (EEAs) typically assume that all pixels lie within a simplex, with the endmembers located at the vertices of this simplex [16]. EEAs based on the pure pixel assumption aim to find the simplex enclosing the hyperspectral data with the largest volume, whose vertices are identified as the endmembers [17], [18], [19]. In contrast, the nonpure pixel

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The authors are with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: fengry@cug.edu.cn; cs.hqpeng01@gmail.com; lizhe.wang@gmail.com).

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assumption-based EEAs estimate endmembers by finding the minimum volume linear simplex that contains the data points [20], [21], [22], [23]. Once the endmembers are determined, the abundance estimation problem transforms into an optimization problem based on the established mathematical model. For LMM, the fully constrained least squares (FCLSs) [24] and sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) [25] are among the most widely applied and effective algorithms for abundance estimation, often incorporating constraints such as the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC).

However, the sequential estimation of endmembers and abundances can lead to error accumulation. To address this, blind unmixing methods, which simultaneously estimate both, have been developed. Common algorithms include those based on nonnegative matrix factorization (NMF) [26], [27], [28], [29], [30] and autoencoders (AEs) [31]. Among these, due to the powerful learning capabilities of deep learning (DL), AE-based methods are increasingly becoming the mainstream approach.

AE-based unmixing methods use an encoder to obtain a low-dimensional representation, from which abundances are estimated under constraints like ANC and ASC. A decoder then reconstructs the image. To the best of our knowledge, the pioneering work by Palsson et al. [31] was the first to propose the use of an AE for blind HU, demonstrating its effectiveness and robustness to noise using various architectural configurations and objective functions. Following this, early AE-based unmixing approaches primarily utilized pixel-wise networks that relied solely on spectral information. In these architectures, the encoder was mainly composed of fully connected layers, and the networks were trained pixel by pixel. Examples of such methods include EndNet [32], uDAS [33], and DAEN [34]. EndNet enhances the AE structure using spectral angle distance (SAD) and a novel loss function combining SAD, Kullback-Leibler (KL) divergence, and sparsity penalties to better capture endmember characteristics. uDAS introduces an untied denoising AE with sparsity, removing the encoder nonnegativity constraint and integrating denoising and an l_{21} constraint to improve robustness against noise and inaccurate endmember number estimation. DAEN employs a deep AE with stacked AEs for robust initialization and a variational AE for simultaneous endmember and abundance estimation, designed to handle outliers and low SNR while enforcing constraints.

The importance of incorporating spatial context for improved unmixing performance quickly became evident. Consequently, convolutional neural networks (CNNs), widely recognized for their ability to capture spatial hierarchies, gained popularity in HU. Zhang et al. [35] proposed an end-to-end CNN-based method for feature extraction and abundance estimation. However, conventional 2-D convolutions primarily operate within the spatial domain and may not fully exploit the rich spectral correlations. To address this, 3-D CNNs were introduced to simultaneously learn both spatial and spectral features. For instance, the work by Zhao et al. [36], [37] explored the use of 3D-CNNs within an AE framework for additive nonlinear unmixing and also proposed a 3D-CNN framework specifically designed to handle spectral variability. Despite their effectiveness, 3-D convolutional kernels often lead to increased parameters and computational complexity. To mitigate this, CrossCUN [38] combined 2-D and 3-D kernels. Furthermore, dual-branch architectures [39] emerged as another solution for separately extracting and fusing spatial and spectral features.

While CNNs effectively capture local spatial features, their inherent inductive bias limits their ability to fully model long-range dependencies and global contextual information across the image. Recently, vision transformers (ViTs), empowered by their self-attention mechanism, have achieved remarkable success in various computer vision tasks by effectively capturing nonlocal feature dependencies [40], [41]. This success has motivated researchers to explore the application of ViT to HU. Among the pioneering efforts, Ghosh et al. [42] applied a transformer network for HU, combining a convolutional AE with a transformer to capture long-range dependencies. Other transformer-based methods have since been proposed. For instance, UnDAT [43] introduced a double-aware transformer designed to simultaneously leverage region homogeneity and spectral correlation through dedicated modules. TCCU-Net [44] proposed a dual-stream collaborative network that combines both transformer and CNN encoders to jointly learn spectral, spatial, global, and local information. Recognizing that ground objects in HSIs often exhibit features at different scales, incorporating multiscale feature extraction into unmixing models is essential. Consequently, several studies have focused on utilizing multiscale spatial information for HU. UST-Net [45] employed a U-shaped transformer network with shifted windows and downsampling/upsampling stages to extract features at different scales across the entire image. MAT-Net [46] proposed a multiscale aggregation Transformer network that uses a dual-stream CNN encoder with branches employing different-sized convolutional kernels to capture multiscale spatial features. Similarly, MSCC-ViT [47] introduced a multiscale visual transformer network that extracts and fuses features at multiple scales by utilizing two different sizes of patches, corresponding to large-scale and small-scale features, along with a convolution crossing attention mechanism.

However, HSIs are inherently 3-D data cubes, encompassing not only spatial but also rich spectral information. While recent works have explored multiscale spatial feature extraction, the potential of multiscale spectral information has often been a focus of research to improve unmixing performance. More recently, the paradigm of jointly exploiting multiscale information from both spectral and spatial domains has gained traction. For instance, Wang et al. [48] proposed a multiscale spectral–spatial network that captures features at different scales and introduced a Boltzmann-inspired adaptive temperature mechanism to refine the abundance estimation process. These works have demonstrated the significant potential of a unified multiscale framework. The central motivation behind our approach stems from the hypothesis that different spectral resolutions provide complementary information essential for robust unmixing. The original, high-resolution

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spectrum is rich in fine-grained diagnostic details, such as narrow-band absorption features, which are crucial for precise material discrimination. However, these high-frequency details are also more susceptible to noise and intraclass spectral variability. Conversely, coarser spectral resolutions, typically obtained through downsampling, effectively suppress noise and preserve the stable, low-frequency characteristics of the spectral continuum. This overall shape represents a more robust, class-level signature of a material. Therefore, an ideal feature representation should synergistically integrate the stable, general features from coarser scales with the discriminative, detailed features from finer scales. While the concept of multiscale analysis is not new, a systematic method to explicitly decompose and hierarchically fuse these complementary features to enhance robustness remains an area for exploration. To address these limitations and fully harness the multidimensional information within HSIs, this article introduces the multiscale spectral–spatial AE network (MS^{2}AE-Net). MS^{2}AE-Net is designed to concurrently extract and integrate multiscale features from both spectral and spatial domains within an AE framework. It achieves this through two specialized components: the MSpecM processes spectral signatures at various resolutions by decomposing them into low-frequency and high-frequency components using fast Fourier transform (FFT). This decomposition is designed to explicitly separate the spectral continuum (captured by low-frequency components) from the fine diagnostic features and noise (captured by high-frequency components). The module then employs adaptive fusion strategies within its core spectral hierarchical fusion block (SHFB) to integrate these complementary features, and the multiscale spatial module (MSpatM), which is constructed from scale-aware spatial blocks (SASBs) that utilize the scale-aware modulation (SAM) concept, incorporating multihead mixed convolution (MHMC) and scale-aware aggregation (SAA) to capture diverse spatial patterns at different scales. By synergistically combining these modules, MS^{2}AE-Net learns a more robust and discriminative representation for HU.

The main contributions of this article are summarized as follows.

1) We propose an MS^{2}AE-Net for HU, which systematically integrates multiscale information from both spectral and spatial domains within a unified framework. By synergistically combining specialized modules for each domain, MS^{2}AE-Net generates more robust and discriminative feature representations, particularly under challenging conditions with noise.
2) We propose the multiscale spectral module (MSpecM), a novel spectral feature extraction module designed to enhance robustness against spectral variability and noise. MSpecM operates by decomposing spectral signals into low-frequency components, corresponding to the spectral continuum, and high-frequency components, which capture fine diagnostic features. This is achieved using FFT, followed by adaptive top-down and bottom-up fusion within an SHFB to learn a robust and discriminative spectral representation.
3) We introduce the MSpatM, a spatial information extraction module built upon SASBs. MSpatM leverages the SAM concept, effectively capturing multiscale spatial features through MHMC and aggregating them using SAA to enhance spatial context understanding.

The remainder of this article is organized as follows. Section II provides the necessary preliminaries for understanding the proposed method. Section III details the proposed MS^{2}AE-Net, including its architecture, multiscale spectral and spatial modules, and loss function. Section IV presents experimental results, covering datasets, metrics, implementation, comparisons, and ablation studies. Finally, Section V concludes the article and suggests future work.

II. Preliminaries

A. Linear Mixture Model

The LMM assumes that the sensor receives light after a single reflection from the material surface. Consequently, the observed spectrum can be represented as a linear combination of the pure spectral signatures of the constituent materials, known as endmembers, where the coefficients of this combination correspond to their respective fractional abundances within the pixel. Mathematically, the LMM can be expressed as

$$\mathbf{Y} = \mathbf{MA} + \mathbf{N} \tag{1}$$

where $\mathbf{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_N] \in \mathbb{R}^{L\times N}$ denotes the original HSI, comprising N pixels and L spectral bands. $\mathbf{M} \in \mathbb{R}^{L\times P}$ is the endmember matrix with P endmembers, and $\mathbf{A}$ is the corresponding abundance matrix. Additive noise is represented by $\mathbf{N} \in \mathbb{R}^{L\times N}$. In addition, to guarantee the physical meaning of abundances, $\mathbf{A}$ is subject to two constraints: ANC and ASC

$$\mathbf{A} \geq \mathbf{0} \tag{2}$$

$$\mathbf{1}_P^T\mathbf{A} = \mathbf{1}_N^T. \tag{3}$$

B. Autoencoder

An AE is an unsupervised neural network architecture, typically composed of an encoder and a decoder, that is widely utilized for learning efficient data representations.

1) *Encoder:* The encoder serves as a feature extractor, mapping the high-dimensional input data Y to a lower dimensional latent representation. This process can be formulated as

$$\hat{\mathbf{A}} = f_e(\mathbf{Y}) \tag{4}$$

where $f_e(\cdot)$ denotes the mapping operation and $\hat{\mathbf{A}}$ denotes estimated abundances. To ensure $\hat{\mathbf{A}}$ satisfy the ANC and ASC, the final layer of the encoder is often a softmax activation function.
2) *Decoder:* The decoder is designed to reconstruct the original input data from the latent representation. For HU tasks, the decoder's structure is typically designed to align with the assumed mixing model. It takes the estimated abundances $\hat{\mathbf{A}}$ and reconstructs the HSI $\hat{\mathbf{Y}}$

$$\hat{\mathbf{Y}} = f_d(\hat{\mathbf{A}}). \tag{5}$$

Specifically, for LMM, the decoder is commonly implemented as a single linear layer without bias. This can be expressed as

$$\hat{\mathbf{Y}} = \mathbf{W}_d\hat{\mathbf{A}}. \tag{6}$$

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In this configuration, the weight matrix $\mathbf{W}_d$ of the linear layer is interpreted as the estimated endmember matrix $\hat{\mathbf{M}}$. To facilitate effective training and convergence, the weights of the decoder are often initialized using endmembers estimated by traditional EEAs.

C. Fast Fourier Transform

The FFT is an efficient algorithm for computing the discrete Fourier transform (DFT). It serves as a fundamental tool for analyzing the frequency components present in a discrete-time signal. Applying FFT to a signal allows us to transform it from its original domain (e.g., time or spatial) to the frequency domain, revealing the distribution of different frequency components and their magnitudes.

In the frequency domain, the low-frequency components represent the overall trend and smooth variations of the signal. Conversely, the high-frequency components capture rapid changes, sharp peaks, and detailed variations. Decomposing a signal into its low and high-frequency parts is a common operation in signal processing for various purposes, including filtering, analysis, and feature extraction.

A signal x can be decomposed into its high-frequency (x_{high}) and low-frequency (x_{low}) components based on their representation in the frequency domain. This process involves computing the DFT, manipulating the frequency spectrum, and then applying the inverse DFT (IDFT).

Given a discrete signal vector x, its DFT is computed as

$$X_f = \mathcal{F}(x) \tag{7}$$

where X_f is the complex-valued frequency spectrum. The magnitude of each frequency component, $|X_{f,k}|$, indicates the strength of the kth frequency.

To isolate high-frequency components, a common approach involves identifying frequency components with significant magnitudes. A threshold T can be determined, for example, based on the magnitudes of the top-k largest frequency components. A filtered frequency spectrum $X_{f,\text{high}}$ is then constructed by setting components with magnitudes less than or equal to T to zero

$$X_{f,\text{high}}[k] = \begin{cases} X_{f,\text{high}}[k], & \text{if } \left|X_{f,\text{high}}[k]\right| > T \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

Applying the IDFT to $X_{f,\text{high}}$ reconstructs the signal's high-frequency component in the original domain

$$x_{\text{high}} = \mathcal{F}^{-1}\left(X_{f,\text{high}}\right). \tag{9}$$

Due to the linear additivity of signals, the low-frequency component x_{low} can be directly calculated by the following formula:

$$x_{\text{low}} = x - x_{\text{high}}. \tag{10}$$

This decomposition effectively separates the signal into parts representing its coarse structure and its fine details.

To provide an intuitive understanding of this decomposition's physical significance, we present a demonstrative example on a Kaolinite spectrum from the United States Geological Survey (USGS) library, as shown in Fig. 1. The original spectrum [see Fig. 1(a)] features a distinct narrow-band doublet absorption valley around 2200 nm, a diagnostic characteristic for this mineral [49]. The decomposition is achieved via a straightforward frequency-based low-pass filter. Specifically, we retain only the lowest 10% of the frequency coefficients from the FFT result to reconstruct the low-frequency component. As shown in Fig. 1(b), this component accurately captures the smooth spectral continuum or envelope. The high-frequency component is then obtained as the residual [see Fig. 1(c)]. It is evident that this high-frequency signal successfully isolates the diagnostic absorption features, with its sharp negative peaks corresponding precisely to the locations of the absorption valleys in the original signal. The values in the high-frequency signal, including the negative peaks, represent the deviation from the continuum baseline, where negative values signify absorption. Thus, our FFT-based decomposition serves as a principled method to disentangle the stable, slowly varying spectral shape from the localized, rapidly changing diagnostic features.

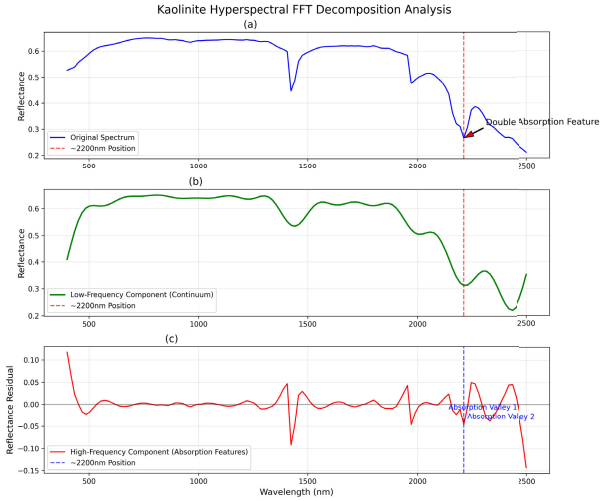


Fig. 1. FFT-based decomposition of a Kaolinite spectrum. (a) Original reflectance signature with a diagnostic absorption doublet near 2200 nm. (b) Extracted low-frequency continuum. (c) Extracted the high-frequency component, isolating the absorption features as a residual signal.

III. Proposed Method

A. Overall Architecture

The overall architecture of the proposed MS^{2}AE-Net is illustrated in Fig. 2. The network follows an AE framework, comprising an encoder and a decoder. The encoder is designed to effectively capture multiscale spectral and spatial features from the input HSI. The data processing flow within the encoder can be described in stages. First, the input HSI $\mathbf{Y} \in \mathbb{R}^{L\times H\times W}$ is processed by the MSpecM to extract multiscale spectral features

$$\mathbf{T} = f_{\text{MSpecM}}(\mathbf{Y}) \tag{11}$$

where $\mathbf{T} \in \mathbb{R}^{C\times H\times W}$ represents the multiscale spectral features extracted by MSpecM. Here, C denotes the number of channels in the intermediate feature map, which is designed

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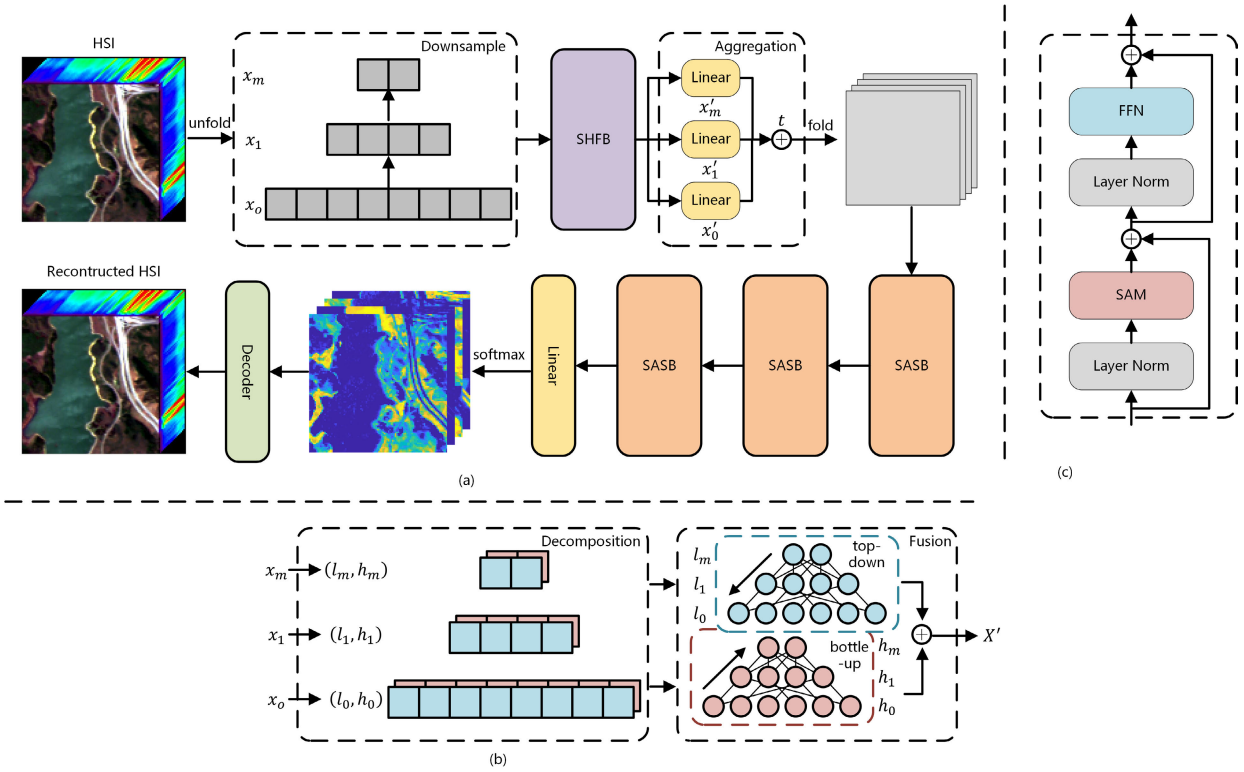


Fig. 2. Proposed MS²AE-Net architecture. (a) Overall framework showing the MSpecM and the MSpatM. (b) Detailed structure of the SHFB, the core of MSpecM. (c) Detailed structure of the SASB, the fundamental unit of MSpatM.

to be smaller than the original number of spectral bands L. This dimensionality reduction after the MSpecM stage effectively compresses the spectral information and contributes to a reduction in the overall number of network parameters in subsequent layers. The MSpecM operates primarily along the spectral dimension, which helps to preserve the spatial structure of the input image. Subsequently, these extracted spectral features $\mathbf{T}$ are fed into the MSpatM to capture multiscale spatial dependencies

$$\mathbf{K} = f_{\text{MSpatM}}(\mathbf{T}). \tag{12}$$

The output $\mathbf{K} \in \mathbb{R}^{P\times H\times W}$ from the MSpatM is then passed through a softmax activation function to obtain the estimated abundance matrix $\hat{\mathbf{A}}$, ensuring the ANC and ASC

$$\hat{\mathbf{A}} = \text{softmax}(\mathbf{S}). \tag{13}$$

The decoder, adhering to LMM, is implemented as a single bias-free linear layer. This layer takes the estimated abundances $\hat{\mathbf{A}}$ and reconstructs the HSI.

B. Multiscale Spectral Module

The MSpecM, inspired by advancements in time-series processing methods [50], [51], is designed to capture spectral features at multiple scales while preserving the spatial structure of the HSI. The MSpecM consists of three main stages: multiscale sequence generation via downsampling, FFT-based decomposition followed by hierarchical feature fusion within the SHFB, and finally, feature aggregation to produce the spectral feature representation. The first step in the MSpecM is to generate multiscale spectral sequences from the input HSI $\mathbf{Y}$. For each pixel i, its spectral vector $\boldsymbol{y}_i \in \mathbb{R}^{1\times L}$ is progressively downsampled to create a set of spectral sequences at different scales, denoted as $X = \{\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_m\}$, where $\boldsymbol{x}_0 = \boldsymbol{y}_i$ is the original spectral vector. The downsampling is performed sequentially, reducing the spectral dimension by a factor of 2 at each step using an average pooling operation

$$\boldsymbol{x}_{k+1} = \text{avgpool}(\boldsymbol{x}_k), \quad k = 0, \ldots, m-1. \tag{14}$$

Next, we adopt the FFT to decompose each spectral sequence $\boldsymbol{x}_k$ at different scales into its high-frequency ($\boldsymbol{h}_k$) and low-frequency ($\boldsymbol{l}_k$) components

$$(\boldsymbol{h}_k, \boldsymbol{l}_k) = f_{\text{FFT}}(\boldsymbol{x}_k), \quad k = 0, \ldots, m \tag{15}$$

where $f_{\text{FFT}}(\cdot)$ represents the FFT-based decomposition process described in Section II. This decomposition separates the invariant (low-frequency) and variant (high-frequency) characteristics of the spectral signal at each scale.

Subsequently, these decomposed low-frequency and high-frequency components across different scales are processed by the core of our spectral feature extraction, the SHFB. The SHFB is designed to effectively fuse these multiscale frequency components. Its computational process to obtain the fused multiscale spectral feature X' is defined as

$$X' = X + \text{FFN}\left(\text{L-Mix}\left(\{\boldsymbol{l}\}_0^m\right) + \text{H-Mix}\left(\{\boldsymbol{h}\}_0^m\right)\right). \tag{16}$$

Here, X represents the original set of multiscale spectral sequences. L-Mix(·) and H-Mix(·) denote the specialized fusion operations within SHFB for the sets of low-frequency

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components $\{\boldsymbol{l}\}_0^m$ and high-frequency components $\{\boldsymbol{h}\}_0^m$ across all scales, respectively. The feed-forward network (FFN), composed of two linear layers and a Gaussian error linear unit (GELU) activation function, further refines these mixed components before they are combined with the original features X via a residual connection.

The fusion strategies L-Mix and H-Mix within the SHFB are designed to leverage the distinct characteristics of the frequency components.

1) Low-Frequency Component Fusion (L-Mix, Top-Down): Low-frequency components represent the macroscopic trends and invariant features of the spectrum. Considering that coarser-scale low-frequency information provides clearer global knowledge, a top-down fusion approach is adopted. Starting from the coarsest scale m, low-frequency information from coarser scales is progressively fused downward layer by layer to guide and enhance the modeling of low-frequency features at the current scale. The fusion process can be expressed as

$$\boldsymbol{l}_k^{\text{mixed}} = \boldsymbol{l}_k + f_{\text{top-down-mixing}}\left(\boldsymbol{l}_{k+1}^{\text{mixed}}\right), \quad k = m-1, \ldots, 0 \tag{17}$$

where $f_{\text{top-down-mixing}}(\cdot)$ is implemented through network layers to achieve adaptive information fusion, consisting of two linear layers and a GELU activation function. $\boldsymbol{l}_k^{\text{mixed}}$ represents the mixed low-frequency component at scale k.

2) High-Frequency Component Fusion (H-Mix, Bottom-Up): High-frequency components contain spectral details and variability information. To fully utilize the detailed information from finer spectral resolutions, a bottom-up fusion approach is employed. Starting from the finest scale 0, high-frequency information from finer scales is progressively integrated upward layer by layer, supplementing detailed information for modeling high-frequency components at coarser scales. The fusion process can be expressed as

$$\boldsymbol{h}_k^{\text{mixed}} = \boldsymbol{h}_k + f_{\text{bottom-up-mixing}}\left(\boldsymbol{h}_{k-1}^{\text{mixed}}\right), \quad k = 1, \ldots, m \tag{18}$$

where $f_{\text{bottom-up-mixing}}(\cdot)$ is implemented through network layers to achieve adaptive information fusion, consisting of two linear layers and a GELU activation function. $\boldsymbol{h}_k^{\text{mixed}}$ represents the mixed high-frequency component at scale k.

Finally, the MSpecM concludes with an aggregation stage. In this stage, linear transformations are applied to the processed features $X' = \{\boldsymbol{x}'_0, \boldsymbol{x}'_1, \ldots, \boldsymbol{x}'_m\}$ from each scale (output by SHFB), and their results are summed to map them to a unified low-dimensional representation. This effectively summarizes the multiscale spectral information into a single vector $\mathbf{t} \in \mathbb{R}^{1\times C}$ for each pixel

$$\boldsymbol{t} = \sum_{k=0}^{m} \left(\mathbf{W}_k \boldsymbol{x}'_k + \boldsymbol{b}_k\right) \tag{19}$$

where $\mathbf{W}_k$ and $\boldsymbol{b}_k$ are learned weight matrices and bias vectors specific to each scale k. This resulting vector $\boldsymbol{t}$ serves as the low-dimensional spectral feature representation for a given pixel, which is then passed to the subsequent spatial processing module.

C. Multiscale Spatial Module

The MSpatM is designed to effectively capture spatial features at multiple scales from the feature maps $\mathbf{T} \in \mathbb{R}^{C\times H\times W}$ (where C is the intermediate channel dimension) generated by the MSpecM. Specifically, the MSpecM processes each pixel's spectrum to extract deep spectral features. These features are then reshaped to form the tensor $\mathbf{T}$, which preserves the original spatial dimensions ($H \times W$). The MSpatM primarily consists of a stack of three SASBs, followed by a final linear layer that projects the features from C dimensions to P dimensions (number of endmembers).

The SASB forms the fundamental building unit of the MSpatM. Its design is analogous to a standard Transformer encoder block, as illustrated in Fig. 2(c). However, the SASB replaces the conventional multihead self-attention (MSA) mechanism with the SAM module. The rest of the block includes layer normalization and an FFN, integrated with residual connections, to process the features. The input to the first SASB is $\mathbf{T}$, and the output of one SASB becomes the input to the next. Let $\mathbf{T}^{(0)} = \mathbf{T}$. The sequential application through the three SASB blocks can be expressed as

$$\begin{aligned}\mathbf{T}^{(1)} &= \text{SASB}_1\left(\mathbf{T}^{(0)}\right)\\ \mathbf{T}^{(2)} &= \text{SASB}_2\left(\mathbf{T}^{(1)}\right)\\ \mathbf{T}^{(3)} &= \text{SASB}_3\left(\mathbf{T}^{(2)}\right).\end{aligned} \tag{20}$$

The final output of the MSpatM before the softmax activation (which is $\mathbf{K}$ in Section III-A) is obtained by passing $\mathbf{T}^{(3)}$ through a linear projection layer: $\mathbf{K} = \mathbf{W}_{\mathbf{proj}}(\mathbf{T}^{(3)})$.

The core of each SASB is the SAM module, inspired by Lin et al. [52]. For an input feature $\mathbf{X}$ to the SAM module, the SAM operation is defined as

$$\begin{aligned}\mathbf{S} &= \text{SAA}(\text{MHMC}(\mathbf{W}_s\mathbf{X}))\\ \mathbf{V} &= \mathbf{W}_v\mathbf{X}\\ \mathbf{Z} &= \mathbf{S}\odot\mathbf{V}\\ \mathbf{X}' &= \mathbf{X}+\mathbf{Z}\end{aligned} \tag{21}$$

where $\odot$ denotes element-wise multiplication. $\mathbf{W}_s$ and $\mathbf{W}_v$ are weight matrices of linear layers that project the input $\mathbf{X}$ to generate the input for MHMC and the value tensor $\mathbf{V}_{\text{value}}$, respectively. The modulator $\mathbf{S}$ changes dynamically with different inputs, enabling adaptive self-modulation. The SAM module itself consists of two main components: MHMC and SAA, shown as Fig. 3.

1) Multihead Mixed Convolution: This component is designed to capture multiscale spatial features and expand the receptive field. It achieves this by partitioning the input channels into multiple "heads" (denoted as n_{head}) and applying distinct depth-wise separable convolutions with varying kernel sizes to each head. For an input feature map $\mathbf{T}$, the MHMC operation can be formulated as follows:

$$\text{MHMC}(\mathbf{T}) = \text{Concat}\left(\text{DW}_{k_1\times k_1}(\boldsymbol{t}_1), \ldots, \text{DW}_{k_n\times k_n}(\boldsymbol{t}_n)\right) \tag{22}$$

where $\boldsymbol{t}_i \in \mathbb{R}^{\lfloor P/n_{\text{head}}\rfloor\times H\times W}$ represents the ith head's portion of the input features, obtained by splitting $\mathbf{T}$ along its channel dimension. $\text{DW}_{k_i\times k_i}(\cdot)$ denotes a depth-wise convolution operation with a kernel size of $k_i \times k_i$. The kernel sizes k_i are varied across heads, starting from 3×3 and increasing by 2 for each subsequent head (e.g., 3×3, 5×5, and 7×7), to capture

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spatial information at diverse granularities. The results from each head are then concatenated (Concat) along the channel dimension to form the output.

2) Scale-Aware Aggregation: The SAA module is designed to effectively aggregate the multigranularity features extracted by the MHMC. It achieves this by shuffling and grouping these multigranularity features along the channel dimension. Specifically, one channel is selected from each head to construct a group. Subsequently, an inverse bottleneck structure is utilized to perform an up-down feature fusion operation within each group, thereby enhancing the diversity of multiscale features. The SAA operation can be formulated as follows:

$$\begin{aligned}\mathbf{H}_j^i &= \mathrm{DW}_{k_j\times k_j}\left(\boldsymbol{t}_j^i\right) \in \mathbb{R}^{1\times H\times W}\\ \mathbf{G}_i &= \mathrm{Conv}_{1\times 1}\left(\left[\mathbf{H}_1^j,\ldots,\mathbf{H}_{n_{\mathrm{head}}}^j\right]\right)\\ \mathbf{K} &= \mathrm{Conv}_{1\times 1}\left(\left[\mathbf{G}_1,\ldots,\mathbf{H}_{n_{\mathrm{group}}}\right]\right)\end{aligned} \tag{23}$$

where $\boldsymbol{H}_j^i$ represents the ith channel slice from the jth head's output feature map (originating from MHMC). $\mathbf{G}_i$ is the ith group's aggregated feature, $n_{\mathrm{group}} = (P/n_{\mathrm{head}})$ signifies the number of groups. The two $\mathrm{Conv}_{1\times1}(\cdot)$ operations perform multigranularity feature aggregation both within groups and globally, effectively forming an inverse bottleneck structure.

D. Loss Function

The objective function for our proposed MS²AE-Net is a composite loss designed to guide the network toward effective HU. It combines the mean squared error (mse) for reconstruction fidelity and the SAD for preserving spectral shape characteristics. The mse component quantifies the average squared difference between the original and reconstructed spectra, thereby measuring the overall reconstruction accuracy. It is defined as

$$\mathcal{L}_{\mathrm{mse}} == \frac{1}{N}\sum_{i=1}^{N}\left\|\boldsymbol{y}_i - \hat{\boldsymbol{y}}_i\right\|^2 \tag{24}$$

where $\boldsymbol{y}_i$ and $\hat{\boldsymbol{y}}_i$ represent the original and reconstructed spectral vectors for the ith pixel, respectively. N is the total number of pixels in the image. While effective for measuring spectral reconstruction fidelity, mse is sensitive to variations in spectral intensity. To address this sensitivity and better preserve the characteristic spectral shapes crucial for distinguishing materials, the SAD metric is incorporated. SAD is a scale-invariant measure that calculates the angle between two spectral vectors. It is given by

$$\mathcal{L}_{\mathrm{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\arccos\left(\frac{\langle \boldsymbol{y}_i, \hat{\boldsymbol{y}}_i\rangle}{\|\boldsymbol{y}_i\|\,\|\hat{\boldsymbol{y}}_i\|}\right). \tag{25}$$

Here, $\langle \boldsymbol{y}_i, \hat{\boldsymbol{y}}_i\rangle$ denotes the dot product between the original spectral vector $\boldsymbol{y}_i$ and the reconstructed spectral vector $\hat{\boldsymbol{y}}_i$, and $\|\cdot\|_2$ represents the Euclidean norm (or L_2 norm) of a vector. To leverage the complementary strengths of both mse and SAD, the total loss function $\mathcal{L}_{\mathrm{total}}$ is formulated as a weighted sum of these two components

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{mse}} + \mu\mathcal{L}_{\mathrm{SAD}}. \tag{26}$$

In this equation, μ is a positive hyperparameter that balances the contributions of the $\mathcal{L}_{\mathrm{mse}}$ and $\mathcal{L}_{\mathrm{SAD}}$ terms. This composite loss function guides the network to achieve robust and accurate spectral unmixing by promoting both precise reconstruction and preservation of spectral characteristics.

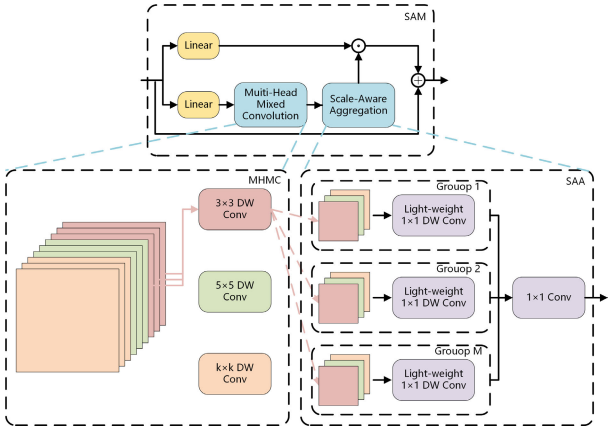


Fig. 3. Detailed architecture of the SAM module. It consists of MHMC for multiscale feature extraction and SAA for feature fusion.

IV. Experiments

A. Datasets

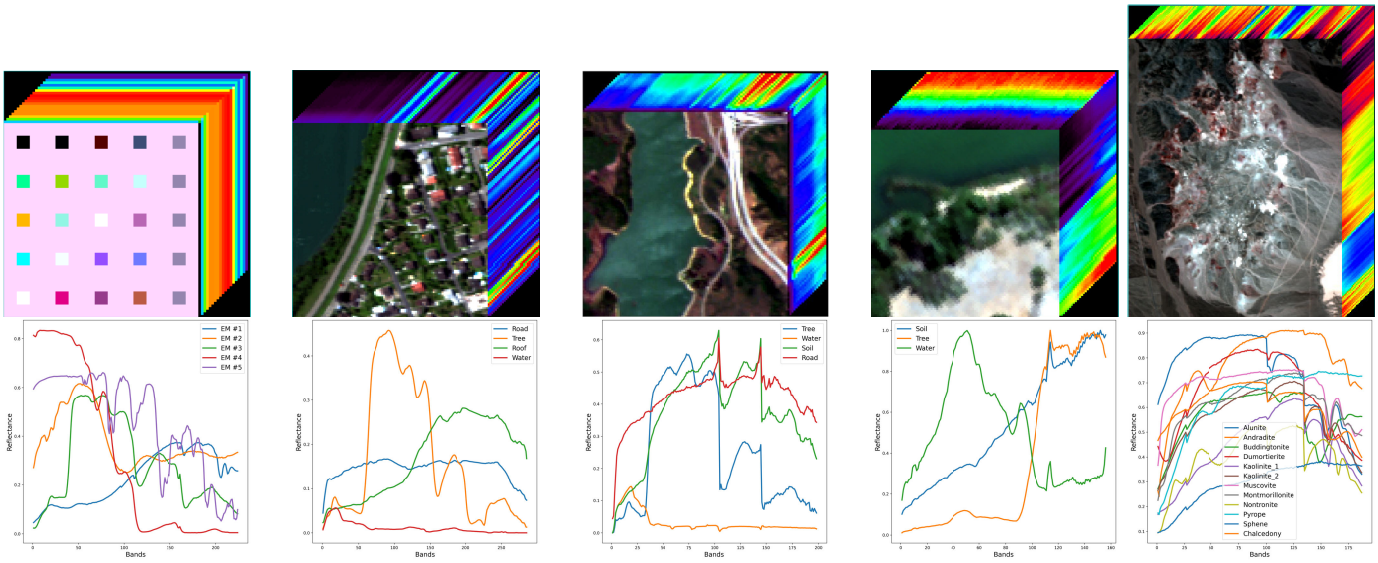
To comprehensively evaluate the performance of our proposed HU network and compare it with state-of-the-art methods, experiments were conducted on one synthetic dataset and three widely used real-world hyperspectral datasets.

1) Synthetic Dataset: This dataset was generated using the LMM. Five reference endmember signatures were randomly selected from the USGS spectral library. The abundance maps employed were derived from the DC1 dataset presented in [53], resulting in an image size of 75×75 pixels. The synthetic HSI comprises 224 spectral bands, covering the wavelength range from 0.4 to 2.5 μm. The false-color representation of the synthetic HSI and its corresponding reference endmembers is illustrated in Fig. 4(a). Furthermore, to assess the robustness of the models under varying noise conditions, Gaussian white noise was added to the synthetic data to achieve signal-to-noise ratios (SNRs) of 30, 20, and 10 dB in separate experimental settings.

2) KU Leuven Ray-Tracer Dataset: To rigorously validate the unmixing performance against ground truth, we also employ the KU Leuven Ray-Tracer Dataset [54]. This dataset is designed to address the validation limitations of real-world imagery by providing a controlled environment with exact reference data. It is generated using a modified physically-based ray-tracer (PBRT) that explicitly models the entire pathway of light rays within 3-D vegetative systems. This simulation inherently captures complex nonlinear mixing events caused by multiple scattering among vegetation components, making it a highly realistic benchmark. In this study, we utilize the "Orchard" scene from this collection. The image has dimensions of 20×20 pixels and comprises two primary endmembers: tree and soil.

3) Apex Dataset: The Apex dataset [55] was acquired by the APEX sensor. The original data cube has dimensions of 1500×1000 pixels and 285 spectral bands, covering the wavelength range from 413 to 2412 nm. Following the removal

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(a) (b) (c) (d) (e)

Fig. 4. False-color HSI and reference endmember of datasets: (a) DC1, (b) Apex, (c) JasperRidge, (d) Samson, and (e) Cuprite.

of bands with low SNR, a subimage of 110 × 110 pixels was selected for the unmixing experiments. The false-color image and reference endmembers for this dataset are shown in Fig. 4(b). This subimage primarily contains four endmembers: road, tree, roof, and water.

4) JasperRidge Dataset: The Jasper Ridge dataset was captured by the AVIRIS sensor. The full scene has dimensions of 512×614 pixels with 224 spectral bands, covering wavelengths from 380 to 2500 nm. After removing bands affected by water vapor absorption and atmospheric interference, a subimage of 100×100 pixels with 198 spectral bands was utilized for experimentation. The false-color image and reference endmembers for this dataset are shown in Fig. 4(c). This region contains four distinct endmembers: tree, water, soil, and road.

5) Samson Dataset: The Samson dataset was acquired by the SAMSON sensor. The original image consists of 952 × 952 pixels and features 156 spectral bands spanning the 401–889-nm wavelength range. For our experiments, a commonly used subimage of 95 × 95 pixels was selected. The false-color image and reference endmembers for this dataset are shown in Fig. 4(d). Three endmembers exist in the Samson dataset: soil, tree, and water.

6) Cuprite Dataset: The Cuprite dataset, captured by the AVIRIS sensor, is a challenging benchmark for unmixing tasks. It originally contains 224 bands ranging from 370 to 2480 nm, with 188 bands remaining after removing noisy and water absorption channels. The experimental subimage covers a 250 × 190 pixel area. Following established literature [48], we consider 12 endmembers due to spectral similarities among mineral variants: "Alunite," "Andradite," "Buddingtonite," "Dumortierite," "Kaolinite1," "Kaolinite2," "Muscovite," "Montmorillonite," "Nontronite," "Pyrope," "Sphene," and "Chalcedony." The false-color image is shown in Fig. 4(e).

B. Evaluation Metrics

To quantitatively assess the performance of the proposed and competing HU algorithms, two widely adopted metrics are employed: the root mean square error (RMSE) for evaluating the accuracy of abundance estimation, and the SAD for measuring the similarity between estimated and reference endmembers.

The RMSE is utilized to quantify the discrepancy between the ground truth abundance matrix $\mathbf{A}$ and the estimated abundance matrix $\hat{\mathbf{A}}$. The RMSE is defined as

$$\text{RMSE}\left(\mathbf{A}, \hat{\mathbf{A}}\right) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\boldsymbol{a}_i - \hat{\boldsymbol{a}}_i\|^2} \tag{27}$$

where $\boldsymbol{a}_i$ and $\hat{\boldsymbol{a}}_i$ represent the true and estimated abundance vectors for the ith pixel, respectively. A lower RMSE value signifies a more accurate abundance estimation, thus indicating superior performance.

For endmember signature similarity, the SAD is utilized. It calculates the spectral angle between the estimated endmembers $\hat{\mathbf{M}}$ and the reference endmembers $\mathbf{M}$, averaged over all endmembers

$$\text{SAD}\left(\mathbf{M}, \hat{\mathbf{M}}\right) = \frac{1}{P}\sum_{i=j}^{P} \arccos\left(\frac{\langle \boldsymbol{m}_j, \hat{\boldsymbol{m}}_j \rangle}{\|\boldsymbol{m}_j\| \, \|\hat{\boldsymbol{m}}_j\|}\right) \tag{28}$$

where $\boldsymbol{m}_j$ and $\hat{\boldsymbol{m}}_j$ are the jth true and estimated endmember spectral signatures, respectively. Smaller SAD values indicate higher similarity between the estimated and reference endmembers, reflecting better extraction performance.

In both metrics, lower values correspond to better unmixing performance.

C. Experimental Setup

1) Network Configuration for MS^{2}AE-Net: The proposed MS^{2}AE-Net was implemented using PyTorch and trained on

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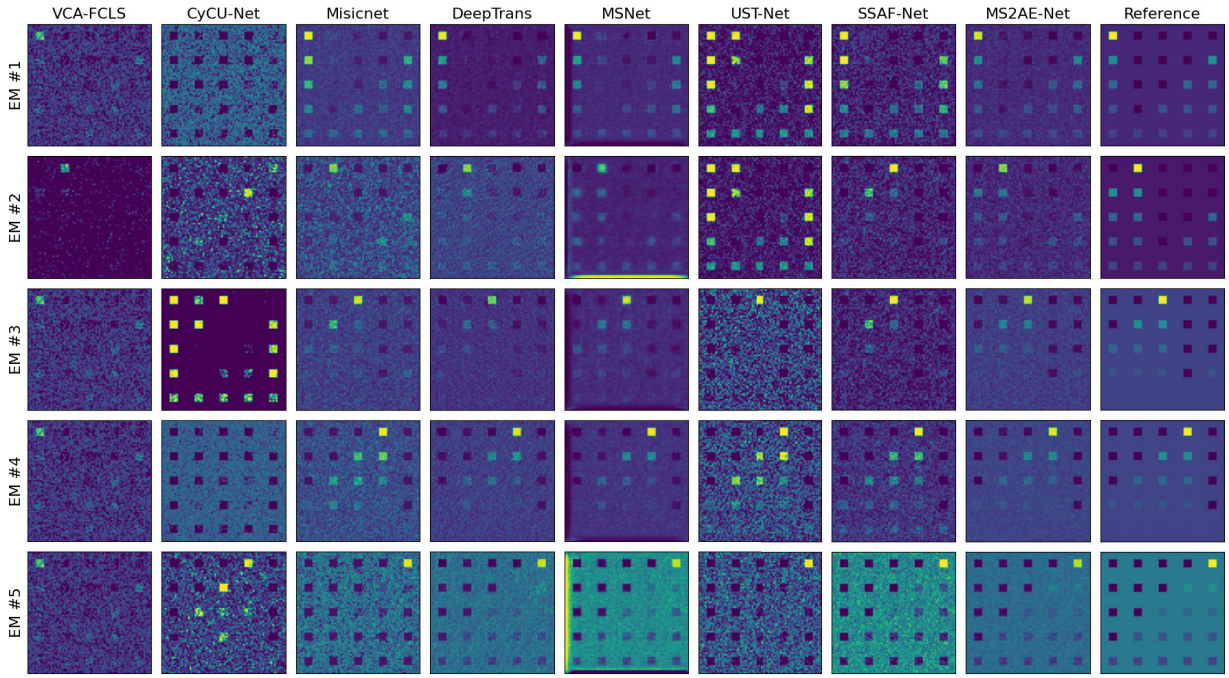


Fig. 5. Comparison of abundance maps from different unmixing algorithms and the reference for the DC1 dataset at 10-dB SNR.

TABLE I
QUANTITATIVE RESULT ON DC1 DATASET WITH DIFFERENT SNRS

Method		VCA-FLCS	CyCU-Net	Misicnet	DeepTrans	MSNet	UST-Net	SSAF-Net	MS2AE-Net
SAD=30dB	SAD	0.3871	0.1946	0.2991	**0.0272**	0.0907	0.2800	0.2481	0.0365
	RMSE	0.1175	0.2039	0.0926	0.0505	0.1159	0.1077	0.1072	**0.0148**
SAD=20dB	SAD	0.4111	0.2068	0.2960	0.0664	0.0878	0.3619	0.2971	**0.0521**
	RMSE	0.1113	0.1998	0.0971	0.0804	0.1168	0.1462	0.1350	**0.0299**
SAD=10dB	SAD	0.5690	0.2384	0.2906	0.0892	0.1075	0.3239	0.2805	**0.0726**
	RMSE	0.1680	0.2376	0.1245	0.0844	0.1035	0.1782	0.1216	**0.0451**

an NVIDIA GeForce RTX 4090 GPU. The Adam optimizer was employed for training with a learning rate set to 0.0002. The network was trained for 500 epochs on the synthetic dataset and 1000 epochs on the three real-world datasets (Apex, JasperRidge, and Samson) to ensure convergence. For the MSpecM, the input spectral sequence was downsampled $m = 3$ times, with the spectral dimension halved at each downsampling stage. We set $m = 3$ as a tradeoff between accuracy and computational cost, since increasing m beyond 3 yields only limited accuracy gains while increasing computation. The final dimensionality C of the extracted spectral features after this module was set to 32. Within the SAM module, the number of heads of MHMC, n_{head}, was configured to 4. All inverted bottleneck structures integrated into our network architecture utilized an expansion ratio of 2 (e.g., a feature map with 32 channels is first expanded to 64 channels and then projected back to 32 channels). The hyperparameter μ balancing the terms in our composite loss function in Section III-D was set to 0.002, which was determined through grid search and demonstrated the best generalization performance across all datasets.

2) Comparative Analysis Setup: To comprehensively evaluate the performance of the proposed MS^{2}AE-Net, we compare it against seven representative unmixing algorithms. These include one traditional method as follows.

1) *VCA-FCLS:* VCA [17] for endmember extraction followed by FCLS [24] for abundance estimation, representing classical approaches based on convex geometry and least squares optimization.

And six DL-based SOTA methods as follows.

1) *CyCU-Net [56]:* A network that primarily focuses on leveraging spatial information for HU.
2) *Misicnet [57]:* This method incorporates spatial and geometrical information by adding a minimum simplex volume penalty to the loss function, designed to be robust even in the absence of pure pixels.
3) *DeepTrans [42]:* Recognized as the first unmixing network to employ a Transformer-based architecture to capture global dependencies.
4) *MSNet [58]:* A multistage convolutional AE network designed to learn broad contextual information and

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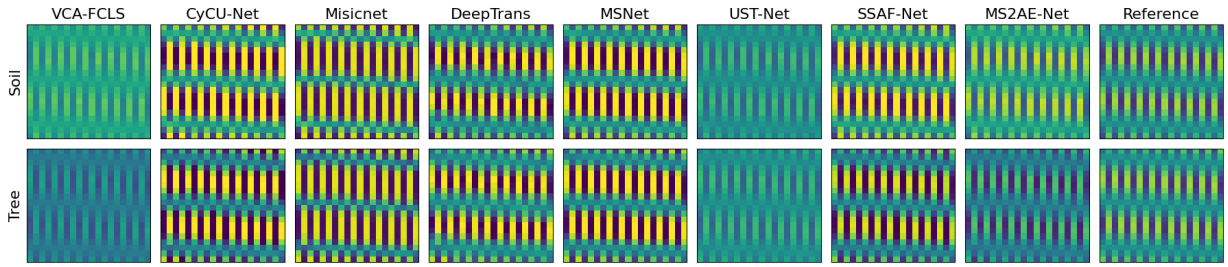


Fig. 6. Estimated abundance maps for soil and tree by different algorithms versus reference maps on the KU Leuven Ray-Tracer Dataset (orchard scene).

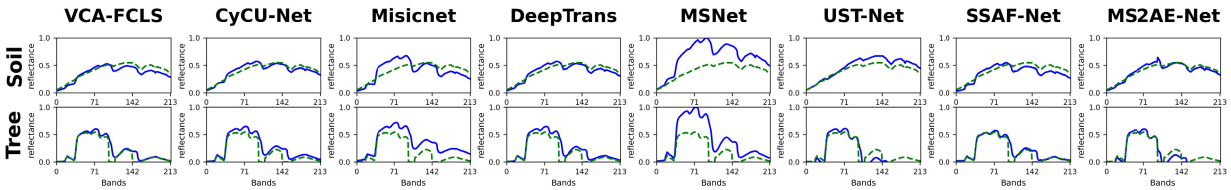


Fig. 7. Estimated endmember spectra (blue solid lines) by different algorithms versus reference spectra (green dashed lines) for soil and tree on the KU Leuven Ray-Tracer Dataset (orchard scene).

long-range dependencies, moving beyond pixel-level associations.

5) *UST-Net [45]:* This U-shaped transformer network uses shifted windows to capture global spatial context and multiscale features effectively from the entire image.
6) *SSAF-Net [59]:* A network that proposes a spatial–spectral adaptive fusion strategy and specifically addresses endmember variability using a deep generative model.

For a fair comparison, the initial endmembers for all DL-based methods, including our proposed MS^{2}AE-Net, were extracted using VCA. The experiments for all comparison algorithms were conducted using their publicly available code. The parameter settings for these methods were adopted directly from the configurations provided within their respective official codebases to ensure reproducibility and adherence to the authors' intended setups. All experiments were conducted under the same hardware and software environment to ensure consistency.

D. Unmixing Experiments on DC1 Dataset

The quantitative unmixing results for the simulated dataset DC1 are presented in Table I, and the corresponding abundance maps at 10 dB are shown in Fig. 5.

From the comprehensive RMSE and SAD metrics in Table I, MS^{2}AE-Net consistently achieves optimal performance, particularly excelling in high-noise scenarios (10, 20 dB) where it surpasses other DL-based methods in both endmember extraction and abundance estimation. The visual results in Fig. 5 corroborate these findings, with MS^{2}AE-Net generating abundance maps that are visibly cleaner and more faithful to the reference compared to other algorithms, especially under the demanding 10-dB noise level.

DeepTrans is a notable competitor, demonstrating comparable performance in both endmember and abundance accuracy. However, MS^{2}AE-Net's design, specifically the MSpecM employing FFT for low and high-frequency component decomposition, provides it with superior robustness against noise. Misicnet shows some strength in abundance estimation due to its deep priors, but its endmember extraction is less competitive. Other methods generally exhibit more noise and lower accuracy in their abundance maps, as shown in Fig. 5.

In summary, both quantitative and visual evaluations confirm that MS^{2}AE-Net delivers state-of-the-art performance, particularly showcasing its robustness in challenging noisy conditions.

E. Unmixing Experiments on KU Leuven Ray-Tracer Dataset

This section presents the unmixing performance on the KU Leuven Ray-Tracer Dataset (orchard scene). The quantitative results are summarized in Table II, while the visual comparisons are illustrated in Figs. 6 and 7.

As shown in Table II, the proposed MS^{2}AE-Net achieves the best overall performance in abundance estimation, recording the lowest mean RMSE of 0.0914. Given that the spatial distribution of the orchard scene is relatively simple, this superior performance is largely attributed to the proposed MSpecM. By decomposing spectra into multiscale frequency components, the MSpecM effectively captures the complex spectral variations caused by nonlinear mixing effects inherent in this physically simulated dataset. Regarding endmember extraction, while VCA-FCLS and UST-Net achieve the lowest SAD values for tree and soil, respectively, MS^{2}AE-Net maintains competitive performance, ranking second for the soil endmember.

Visually, as observed in Fig. 6, a distinct trend emerges. With the exception of VCA-FCLS, UST-Net, and MS^{2}AE-Net, other competing methods tend to oversimplify the scene by identifying mixed pixels as pure materials. This indicates a weaker capability in handling highly mixed scenarios. In contrast, MS^{2}AE-Net accurately preserves the

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TABLE II

QUANTITATIVE COMPARISONS ON THE KU LEUVEN RAY-TRACER DATASET (ORCHARD SCENE). THE BEST AND SECOND-BEST RESULTS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

Methods		VCA-FCLS	CyCU-Net	Misicnet	DeepTrans	MSNet	UST-Net	SSAF-Net	MS2AE-Net
SAD	Soil	0.1224	0.1153	0.2797	0.1278	0.1674	**0.0667**	0.1724	<u>0.0948</u>
	Tree	**0.2578**	0.2804	0.3959	<u>0.2697</u>	0.2820	0.4374	0.2733	0.3066
	Mean	**0.1901**	<u>0.1979</u>	0.3378	0.1987	0.2247	0.2520	0.2228	0.2007
RMSE	Soil	0.1571	0.1469	0.1619	<u>0.0984</u>	0.1320	0.1130	0.1149	**0.0914**
	Tree	0.1571	0.1469	0.1619	<u>0.0984</u>	0.1320	0.1130	0.1149	**0.0914**
	Mean	0.1571	0.1469	0.1619	<u>0.0984</u>	0.1320	0.1130	0.1149	**0.0914**

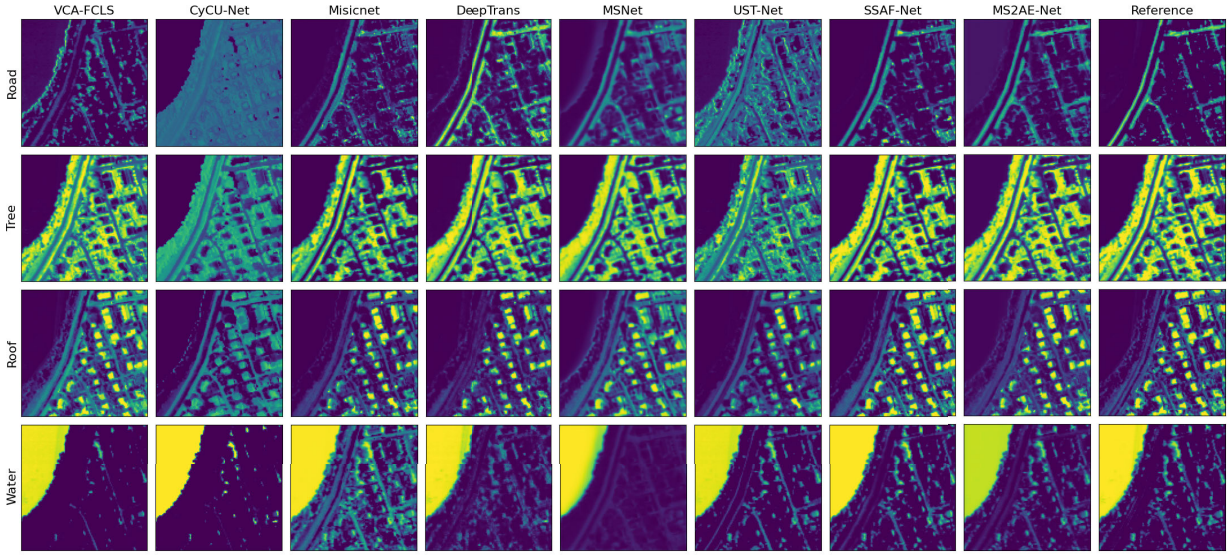


Fig. 8. Estimated abundance maps for road, tree, roof, and water by different algorithms versus reference maps on the Apex dataset.

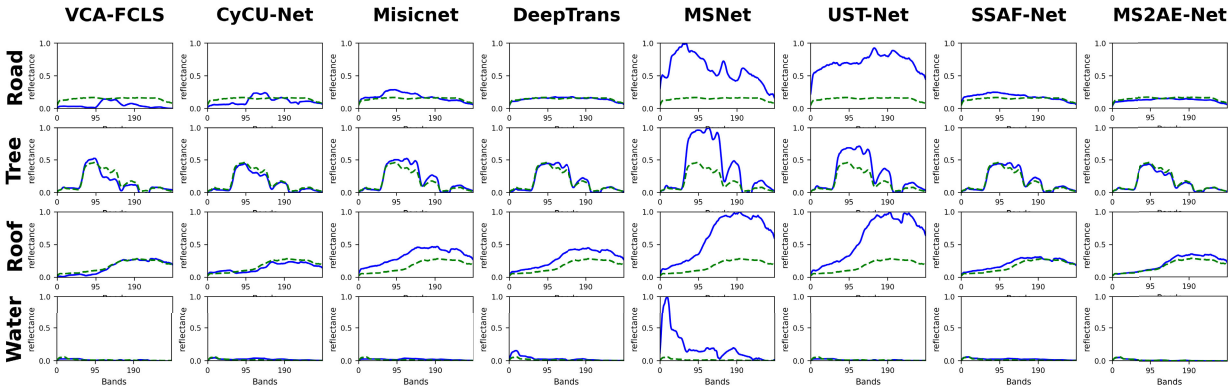


Fig. 9. Estimated endmember spectra (blue solid lines) by different algorithms versus reference spectra (green dashed lines) for road, tree, roof, and water on the Apex dataset.

gradations of the mixture, demonstrating robust unmixing capabilities.

F. Unmixing Experiments on the Apex Dataset

Quantitative unmixing results for the Apex dataset, including RMSE and SAD metrics, are presented in Table III. Fig. 8 displays the abundance maps estimated by the various algorithms, while Fig. 9 compares the extracted endmember spectra against the reference spectra. The results in Table III indicate that our proposed MS^{2}AE-Net achieves the most favorable overall unmixing performance on the Apex dataset, securing the lowest mean SAD and mean RMSE values.

In terms of abundance estimation, MS^{2}AE-Net demonstrated leading RMSE performance for several endmembers and the best overall mean RMSE. As illustrated in Fig. 8, the abundance maps generated by MS^{2}AE-Net show a strong correspondence with the reference maps, exhibiting clear material delineations. This can be attributed to the efficacy of our proposed MSpatM in capturing relevant spatial context. SSAF-Net also showed competitive abundance results, and MSNet performed commendably. Conversely,

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TABLE III

QUANTITATIVE COMPARISONS ON THE APEX DATASET. THE BEST AND SECOND-BEST RESULTS FOR EACH ENDMEMBER TYPE AND THE MEAN ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

Methods		VCA-FCLS	CyCU-Net	Misicnet	DeepTrans	MSNet	UST-Net	SSAF-Net	MS2AE-Net
SAD	Road	0.6915	0.4543	0.2705	0.0836	0.2766	0.0907	0.1909	**0.0824**
	Tree	0.2644	0.0850	0.1375	0.1295	0.1378	0.1339	0.1285	**0.0445**
	Roof	0.1471	0.1298	0.1898	0.0903	0.0817	0.0689	0.1548	**0.0515**
	Water	0.5176	0.6223	0.7451	0.0434	**0.0383**	0.5040	0.4794	0.0646
	Mean	0.4052	0.3228	0.3357	0.0867	0.1336	0.1994	0.2384	**0.0607**
RMSE	Road	0.2331	0.2921	0.1322	0.1776	0.1434	0.3078	0.1210	**0.0979**
	Tree	0.0944	0.2020	0.1458	0.0993	0.1411	0.1907	0.0959	**0.0462**
	Roof	0.1201	0.1630	0.1059	0.1200	0.1143	0.1483	**0.0775**	0.0843
	Water	0.1327	0.1213	0.1842	0.0902	0.1537	0.0797	**0.0386**	0.0625
	Mean	0.1543	0.2046	0.1448	0.1264	0.1389	0.1997	0.0885	**0.0754**

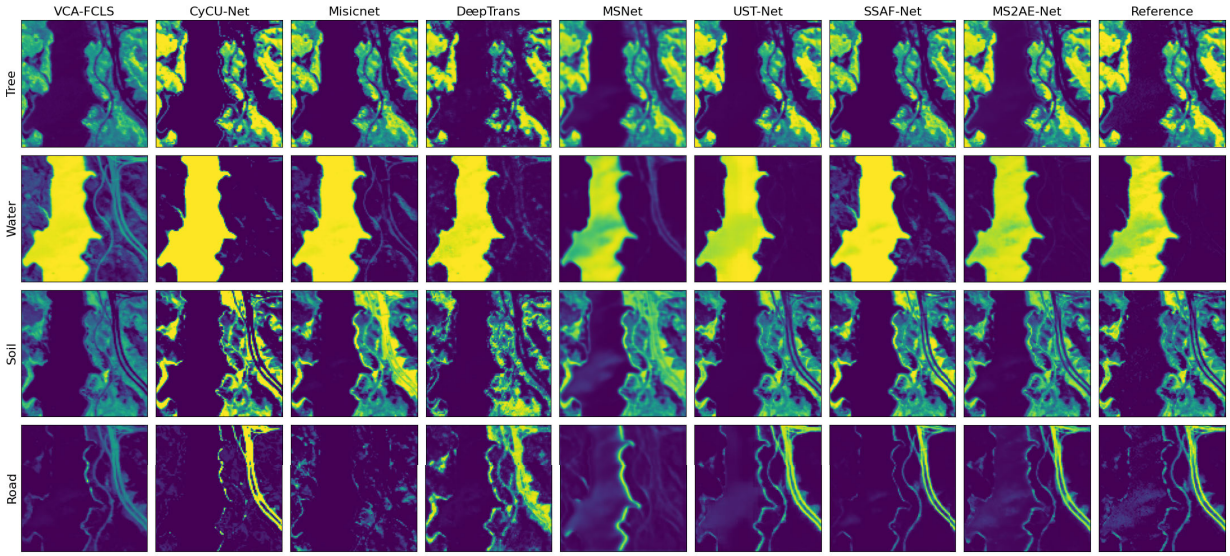


Fig. 10. Estimated abundance maps for tree, water, soil, and road by different algorithms versus reference maps on the JasperRidge dataset.

TABLE IV

QUANTITATIVE COMPARISONS ON THE JASPERRIDGE DATASET. THE BEST AND SECOND-BEST RESULTS FOR EACH ENDMEMBER TYPE AND THE MEAN ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

Methods		VCA-FCLS	CyCU-Net	Misicnet	DeepTrans	MSNet	UST-Net	SSAF-Net	MS2AE-Net
SAD	Tree	0.1481	0.0552	0.0434	0.0291	0.0901	0.0685	0.0442	**0.0214**
	Water	0.1051	0.1627	0.2897	0.0364	0.0252	0.0292	0.2453	**0.0000**
	Soil	0.1166	0.0680	0.0663	0.1825	0.1016	0.0395	0.0246	**0.0134**
	Road	0.0901	0.1044	0.3296	0.1147	0.8942	0.0439	**0.0280**	0.0370
	Mean	0.1150	0.0976	0.1822	0.0907	0.2778	0.0453	0.0855	**0.0179**
RMSE	Tree	0.1583	0.0982	0.0957	0.1827	0.0757	0.0597	0.0979	**0.0399**
	Water	0.2047	0.1009	0.1144	0.0820	0.1259	0.0369	0.1160	**0.0276**
	Soil	0.1311	0.1429	0.2212	0.2604	0.1799	0.0853	0.0963	**0.0776**
	Road	0.1143	0.1179	0.2489	0.2270	0.2443	0.0798	0.0783	**0.0572**
	Mean	0.1559	0.1163	0.1824	0.1996	0.1685	0.0681	0.0980	**0.0539**

UST-Net's performance was impacted by some misclassification between certain endmembers, visible in its abundance maps.

For endmember spectra extraction, MS^{2}AE-Net obtained the lowest SAD values for the majority of endmembers, indicating a high fidelity in capturing the true spectral signatures, as also supported by the visual comparisons in Fig. 9. This enhanced spectral discrimination is a direct benefit of our novel MSpecM, which excels at extracting detailed spectral characteristics. DeepTrans also demonstrated strong competitiveness in endmember estimation.

G. Unmixing Experiments on the JasperRidge Dataset

The unmixing performance of various algorithms on the JasperRidge dataset is presented in this section. Visual comparisons of the estimated abundance maps and extracted endmember spectra are provided in Figs. 10 and 11, respectively. The quantitative metrics, including SAD and RMSE, are summarized in Table IV. Based on a comprehensive evaluation of these results, our proposed MS^{2}AE-Net achieved the most optimal unmixing performance overall on this dataset.

Several other methods also demonstrated notable characteristics. UST-Net, for example, showed strong competitiveness,

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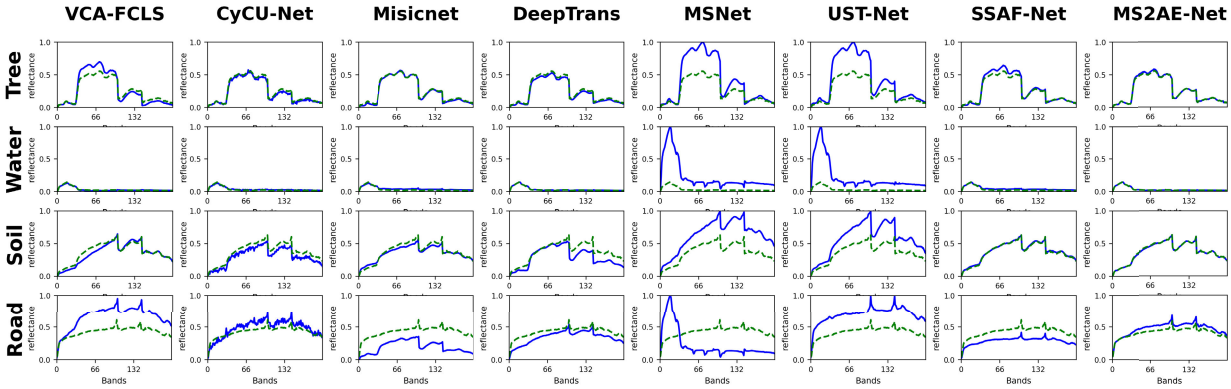


Fig. 11. Estimated endmember spectra (blue solid lines) by different algorithms versus reference spectra (green dashed lines) for tree, water, soil, and road on the JasperRidge dataset.

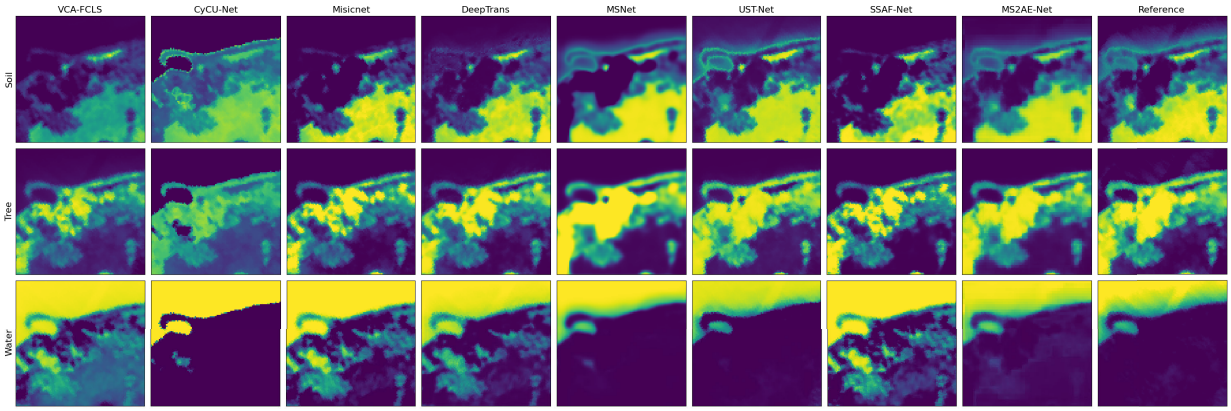


Fig. 12. Estimated abundance maps for soil, tree, and water by different algorithms versus reference maps on the Samson dataset.

a result that can be attributed to its effective multiscale spatial information extraction capabilities. Similarly, SSAF-Net delivered a commendable performance, benefiting from its proficiency in extracting spatial–spectral information. Regarding CyCU-Net, its primary limitation was observed in abundance estimation, where the imposition of an overly strong sparsity constraint led to excessively sparse abundance maps, not fully reflecting the ground truth distributions. Besides, some models faced specific challenges. DeepTrans exhibited misclassification between the "road" and "soil" endmembers. This is a frequently encountered issue on the JasperRidge dataset, partly due to the inherent spectral similarity between these two materials, which can be observed in the endmember spectra plots (see Fig. 11). Furthermore, a common challenge for many DL-based models is their reliance on VCA for endmember initialization. On this particular dataset, VCA tends to perform suboptimally in extracting the "road" endmember, consequently impacting the unmixing accuracy of models dependent on it. In contrast, our proposed MS^{2}AE-Net, particularly due to the deep multiscale spectral information extraction capability of its MSpecM, demonstrated robust performance and effectively distinguished these spectrally similar endmembers.

In summary, the experimental evaluation on the JasperRidge dataset highlights the superior unmixing capabilities of the proposed MS^{2}AE-Net. Its ability to effectively address known dataset-specific challenges, such as the accurate discrimination of spectrally similar endmembers like "road" and "soil" through the strengths of its MSpecM, and its overall strong quantitative and qualitative results, establishes MS^{2}AE-Net as the best performing model among the compared methods for this dataset.

H. Unmixing Experiments on the Samson Dataset

This section evaluates the unmixing performance of the compared algorithms on the Samson dataset. Figs. 12 and 13 provide visual comparisons of the estimated abundance maps and extracted endmember spectra, respectively. Quantitative metrics, including SAD and RMSE, are detailed in Table V. On this particular dataset, our proposed MS^{2}AE-Net did not achieve the overall optimal performance, though it demonstrated competitive aspects in abundance estimation.

Analyzing the abundance estimation results, while MS^{2}AE-Net was not the top-performing method, its results were notably close to those of the leading methods, UST-Net (optimal) and MSNet (second-best). This is a significant observation, especially considering that MS^{2}AE-Net's endmember estimation on this dataset was not ideal, which led to some minor misclassifications, particularly in the "water" abundance map as shown in Fig. 12. The ability of MS^{2}AE-Net to maintain competitive abundance accuracy

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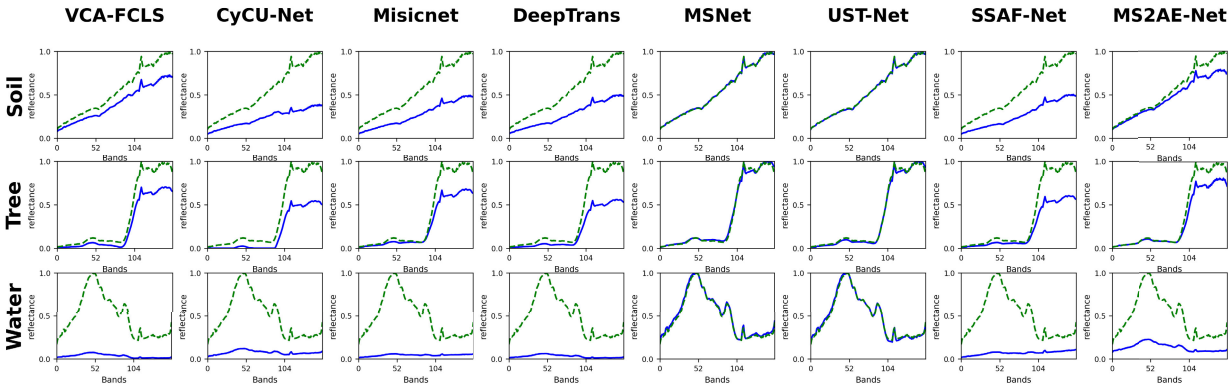


Fig. 13. Estimated endmember spectra (blue solid lines) by different algorithms versus reference spectra (green dashed lines) for soil, tree, and water on the Samson dataset.

TABLE V

QUANTITATIVE COMPARISONS ON THE SAMSON DATASET. THE BEST AND SECOND-BEST RESULTS FOR EACH ENDMEMBER TYPE AND THE MEAN ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY

Methods		VCA-FCLS	CyCU-Net	Misicnet	DeepTrans	MSNet	UST-Net	SSAF-Net	MS2AE-Net
SAD	Soil	0.0207	0.1085	0.0103	**0.0078**	0.0079	0.0091	0.0264	0.0535
	Tree	0.0495	0.1038	0.0354	0.0370	0.0371	0.0332	**0.0310**	0.0413
	Water	0.1299	0.2352	0.4005	0.0479	**0.0250**	0.0276	0.5335	0.1417
	Mean	0.0667	0.1492	0.1487	0.0309	0.0233	**0.0233**	0.1970	0.0788
RMSE	Soil	0.2566	0.1941	0.1819	0.1484	0.0767	**0.0654**	0.1981	0.0665
	Tree	0.1821	0.1660	0.1790	0.1261	0.0659	**0.0510**	0.1764	0.0643
	Water	0.3489	0.1563	0.3131	0.2335	**0.0320**	0.0537	0.3167	0.0626
	Mean	0.2712	0.1729	0.2332	0.1756	0.0582	**0.0567**	0.2385	0.0645

despite inaccuracies in endmember extraction can be largely credited to the robust multiscale spatial information extraction capability of its MSpatM. This module effectively leverages spatial context to refine abundance estimates.

Comparing the top performers, UST-Net and MSNet, both exhibited strong results. While their endmember estimation accuracies were comparable (with UST-Net holding a marginal advantage in mean SAD before rounding to four decimal places as indicated in Table V), UST-Net achieved superior abundance estimation results. A potential reason for UST-Net's enhanced performance in abundance mapping, given similar endmember quality, could be its U-shape architecture, which typically allows for a larger receptive field, thereby facilitating a more comprehensive understanding of global spatial distributions.

In summary, while MS^{2}AE-Net did not lead on the Samson dataset primarily due to challenges in endmember extraction, its performance in abundance estimation remained highly competitive. This underscores the significant contribution of its MSpatM in effectively processing spatial information, even when faced with less than ideal spectral inputs.

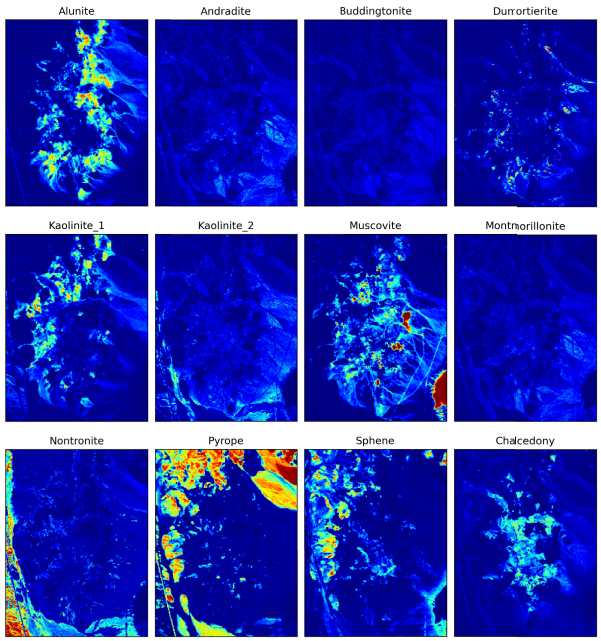


Fig. 14. Estimated abundance maps for the 12 minerals on the Cuprite dataset obtained by MS^{2}AE-Net.

I. Unmixing Experiments on the Cuprite Dataset

This section presents the unmixing results on the Cuprite dataset. As previously mentioned, this dataset is characterized by significant spectral variability and lacks reliable ground truth for abundance maps. Due to these challenges, most comparison algorithms failed to yield consistent or physically meaningful unmixing results in our experiments. Consequently, we focus on a visual analysis of the results obtained by the proposed MS^{2}AE-Net.

Figs. 14 and 15 display the estimated endmember spectra and the corresponding abundance maps, respectively. To align the estimated endmembers with the reference library signatures, we employed the Hungarian algorithm to minimize

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TABLE VI

ABLATION STUDY OF DIFFERENT MODULE COMBINATIONS ON THE APEX, JASPERRIDGE, AND SAMSON DATASETS. "✓" DENOTES THE COMPONENT/STRATEGY IS ACTIVE, "×" DENOTES IT IS INACTIVE. BEST RESULTS ARE HIGHLIGHTED IN BOLD

Spectral Path	MSpecM	Spatial Path	MSpatM	Apex		JasperRidge		Samson	
				SAD	RMSE	SAD	RMSE	SAD	RMSE
✓	×	×	×	0.1419	0.1529	0.1559	0.1527	0.2070	0.2229
✓	✓	×	×	0.0998	0.1085	0.0925	0.1048	0.1511	0.1732
×	×	✓	×	0.1652	0.1439	0.1645	0.1372	0.2493	0.1899
×	×	✓	✓	0.1033	0.1158	0.0911	0.0987	0.1205	0.1055
✓	×	✓	×	0.1250	0.1301	0.1322	0.1258	0.1853	0.1714
✓	×	✓	✓	0.0751	0.0869	0.0488	0.0723	0.0903	0.0712
✓	✓	✓	×	0.0724	0.0891	0.0315	0.0650	0.1156	0.1097
✓	✓	✓	✓	**0.0607**	**0.0754**	**0.0179**	**0.0539**	**0.0788**	**0.0645**

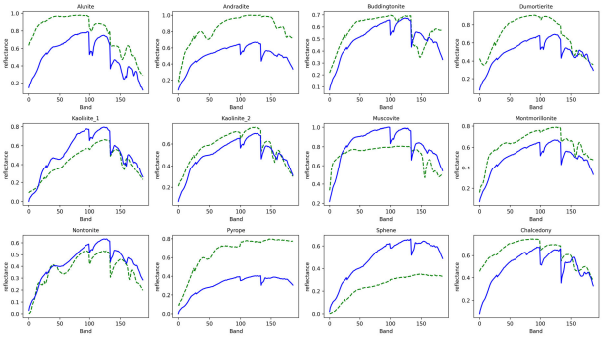


Fig. 15. Comparison between the estimated endmember spectra (blue solid lines) by MS^{2}AE-Net and the reference library spectra (green dashed lines) for the Cuprite dataset.

the total SAD. The resulting alignment yields an average SAD of 0.1150, demonstrating that MS^{2}AE-Net can effectively recover the diagnostic spectral features of the minerals despite the complex interference. The abundance maps in Fig. 14 are plotted based on this correspondence, showing clear and coherent distribution patterns for the 12 extracted minerals.

J. Ablation Study

To validate the effectiveness of the proposed MSpecM and MSpatM, we conducted a comprehensive ablation study on the Apex, JasperRidge, and Samson datasets. The results, presented in Table VI, systematically evaluate the network's performance under various configurations.

In the table, "✓" signifies an active processing path or the use of a specific multiscale module, while "×" indicates its absence. Specifically, when a "spectral path" is active (✓) but "MSpecM" is inactive (×), basic spectral processing using stacked 1×1 convolutions is employed. Similarly, an active "spatial path" (✓) without an active "MSpatM" (×) utilizes basic spatial processing with stacked 3×3 convolutions; such approaches for basic spectral and spatial feature extraction, respectively, are also noted in literature [60]. The use of our proposed MSpecM or MSpatM is indicated when their respective columns are marked with ✓ (which also implies their corresponding "path" is active). The configurations evaluated include: 1) using only a basic spectral or spatial processing path; 2) incorporating MSpecM or MSpatM individually, replacing their basic counterparts; 3) combining a basic processing path (e.g., basic spectral) with the proposed multiscale module of the other type (e.g., MSpatM); and 4) the full model employing both MSpecM and MSpatM.

As demonstrated in Table VI, integrating either the MSpecM or MSpatM (indicated by "✓" in their respective columns) generally leads to improved performance over configurations relying solely on basic convolutional blocks or having no corresponding processing path. For instance, comparing the model with only a basic spectral path (row 1: spectral path ✓, MSpecM ×) to one incorporating MSpecM (row 2: spectral path ✓, MSpecM ✓), a clear enhancement in both SAD and RMSE metrics is observed across all datasets. A similar trend is noticeable when introducing MSpatM for the spatial path.

In summary, the final row of Table VI, representing the complete MS^{2}AE-Net architecture where both "MSpecM" and "MSpatM" are employed, consistently achieves the best SAD and RMSE values across all three datasets. This demonstrates the synergistic benefits and individual contributions of both proposed multiscale modules in enhancing the overall unmixing performance.

V. CONCLUSION

This article introduced the MS^{2}AE-Net, a DL framework that integrates two innovative components designed to enhance HU performance by comprehensively exploiting multiscale information from both spectral and spatial domains. The core of MS^{2}AE-Net lies in its two innovative components: the MSpecM and the MSpatM. The MSpecM effectively captures diverse spectral characteristics by decomposing spectral signals into frequency components using FFT and fusing them hierarchically, allowing for better discrimination of invariant and variant spectral features across different scales. Distinct from traditional wavelet-based approaches that typically utilize transforms for static feature extraction, our design integrates FFT decomposition into an end-to-end learnable framework. This choice is motivated by the computational efficiency of FFT and its direct physical interpretation in separating the stable spectral continuum from fine diagnostic features, thereby facilitating adaptive feature fusion guided by the unmixing objective. The MSpatM leverages the SAM concept, employing MHMC and SAA within its SASBs, to extract rich multiscale spatial contextual information. Extensive experiments conducted on one synthetic and three real-world hyperspectral datasets validated the effectiveness and superiority of the proposed MS^{2}AE-Net compared to several state-of-the-art unmixing techniques. The results consistently

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demonstrated MS^{2}AE-Net's ability to achieve lower RMSE in abundance estimation and competitive SAD in endmember extraction, showcasing its robustness, especially in noisy conditions and complex scenes. Ablation studies further confirmed the significant individual and synergistic contributions of the MSpecM and MSpatM to the overall network performance.

While MS^{2}AE-Net shows strong performance, particularly in abundance estimation, future work could focus on further enhancing endmember extraction accuracy, possibly by integrating more advanced endmember modeling techniques or end-to-end learnable endmember dictionaries. In addition, exploring adaptive mechanisms for balancing the contributions of spectral and spatial information across different datasets and investigating lighter network architectures for improved computational efficiency present promising avenues for future research. Finally, regarding dataset selection, we recognize that incorporating advanced physically based simulation datasets and standard mineralogical benchmarks (e.g., the Cuprite scene) would provide more challenging and credible baselines for future methodological comparisons.

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Ruyi Feng (Senior Member, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2011, and the M.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2013 and 2016, respectively.

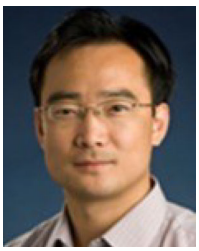
She has been with the School of Computer Science, China University of Geosciences, Wuhan, since 2016, where she is currently an Associate Professor. She is the PI of two National Natural Science Foundation of China (NSFC) projects. Her research interests include sparse representation, deep learning, hyperspectral image analysis, high-resolution remote sensing understanding, and intelligent interpretation of remote sensing imagery.

Dr. Feng is an Associated Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, an Editorial Board Member of *Frontiers in Remote Sensing*, and a Guest Editor of *Remote Sensing*. She served as a reviewer for more than ten SCI-indexed journals.



Hangqi Peng (Student Member, IEEE) received the B.S. degree from China University of Geosciences, Wuhan, China, in 2023, where he is currently pursuing the M.S. degree in computer technology.

His research interests include hyperspectral image processing, hyperspectral unmixing, and deep learning.



Lizhe Wang (Fellow, IEEE) received the B.S. and M.S. degrees from Tsinghua University, Beijing, China, in 1998 and 2001, respectively, and the D.E. degree (Magna Cum Laude) from the University of Karlsruhe, Karlsruhe, Germany, in 2007.

He is the Dean and the "ChuTian" Chair Professor with the School of Computer Science, China University of Geosciences, Wuhan, China. His research interests include remote sensing data processing, digital Earth, and big data computing.

Dr. Wang is a fellow of the Institution of Engineering and Technology (IET) and the British Computer Society (BCS). He was a recipient of the Distinguished Young Scholars of NSFC, the National Leading Talents of Science and Technology Innovation, and the 100-Talents Program of Chinese Academy of Sciences. He was an Associate Editor of *Remote Sensing* and IJDE. He is an Associate Editor of *Remote Sensing*, the *International Journal of Digital Earth*, *ACM Computing Surveys*, IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, and IEEE TRANSACTIONS ON SUSTAINABLE COMPUTING.

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